NAND to Tetris-I

Lab 1: Digital Logic & ALU Design

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Q1. Q-3 (5 marks)

chip Half_Adder (IN a ,b ; OUT sum, carry: Parts: Xor(a=a ,b=b, out =sum); And(a=a, b=b, out = carry): chip full_Adder (IN a ,b, c: out sum , carry: Parts: Half_Adder(a=a ,b=b , sum = s1, carry = c1); Half_Adder(a=s1 , b=c, sum = sum, carry = c2); or(a=c1, b=c2, out=carry);)

Q2. Q-4 (5 marks)

chip Add16{ IN a(16) ,b(16) out out(16); Parts: HalfAdder(a=a(0),b=b(0), sum= out(0) , carry =c(0); Full_Adder(a=a1,b=b1,c=c0), sum=out(1), carry=c(1); Full_Adder(a=a2,b=b2,c=c1), sum=out(2),carry=c(2); Full_Adder(a=a3,b=b3,c=c2), sum=out(3),carry=c(3); Full_Adder(a=a4,b=b4,c=c3), sum=out(4),carry=c(4); Full_Adder(a=a5,b=b5,c=c4), sum=out(5),carry=c(5); Full_Adder(a=a6,b=b6,c=c5), sum=out(6),carry=c(6); Full_Adder(a=a7,b=b7,c=c6), sum=out(7),carry=c(7); Full_Adder(a=a8,b=b8,c=c7), sum=out(8),carry=c(8); Full_Adder(a=a9,b=b9,c=c8), sum=out(9),carry=c(9); Full_Adder(a=a10,b=b10,c=c9), sum=out(10),carry=c(10); Full_Adder(a=a11,b=11,c=c10), sum=out(11),carry=c(11); Full_Adder(a=a12,b=b12,c=c11), sum=out(12),carry=c(12); full_adder(a=a13,b=b13,c=c12), sum=out(13),carry=c(13); full_adder(a=a14,b=b14,c=c13), sum=out(14),carry=c(14); full_adder(a=a15,b=b15,c=c14), sum=out(15),carry=drop); }